STATISTICS USING R PROGRAMMING

Total Marks: 50

Time: 2 Hours

Question 1 [8 marks]

Data Visualization in R

1. (a) Load the **`mtcars`** dataset available in R and plot the following:

    - A scatter plot of `mpg` (miles per gallon) vs `hp` (horsepower).
    - A box plot of `mpg` for each category of `cyl` (cylinders).

    Provide the R code and corresponding plots.
    (4 marks)

2. (b) Using the same dataset, create a **histogram** of the `mpg` variable with appropriate labeling and title.
    (4 marks)

Question 2 [6 marks]

Dataframes and Lists in R

1. (a) Create a dataframe that includes the following information for 5 individuals:

    - Name (character)
    - Age (numeric)
    - Gender (factor)
    - Height (numeric)

    Display the dataframe using appropriate R functions.
    (3 marks)

2. (b) Convert the dataframe you created in part (a) into a **list**. Show the structure of the list and extract the **Age** column from the list.
    (3 marks)

Question 3 [6 marks]

Matrices and Arrays in R

1. (a) Create a **matrix** of size 3×4 filled with the numbers from 1 to 12. Name the rows and columns appropriately and display the matrix.
    (3 marks)

2. (b) Create a **3D array** of dimensions 2×3×4, filled with numbers from 1 to 24. Display the structure of the array.
   (3 marks)

Question 4 [8 marks]

Reading and Extracting Data from Files

1. (a) Assume you have a CSV file named **`sales_data.csv`** with the following columns: `Salesman`, `Region`, `Sales`. Write the R code to:
   - Read the file into a dataframe.
   - Extract and display all rows where `Sales` exceed 5000.
     (4 marks)
2. (b) Suppose you have a text file named **`grades.txt`** containing student names and their grades. Write the R code to:
   - Read the file using appropriate R functions.
   - Convert the data into a dataframe.
     (4 marks)

Question 5 [10 marks]

Linear Regression in R

1. (a) Load the **`mtcars`** dataset again and perform a **linear regression** to predict `mpg` using the variables `wt` (weight) and `hp` (horsepower). Provide the R code to:
   - Fit the model using the `lm()` function.
   - Display the summary of the regression model.
     (5 marks)
2. (b) Plot the regression line obtained from the model for `mpg` vs `wt`.
   (2 marks)
3. (c) Write the R code to predict `mpg` when `wt = 3` and `hp = 100` using the regression model.
   (3 marks)

Question 6 [12 marks]

Correlation in R

1. (a) Load the **`iris`** dataset and calculate the following correlation coefficients for the variables `Sepal.Length` and `Petal.Length`:
   - Pearson correlation coefficient.
   - Spearman correlation coefficient.

- Kendall correlation coefficient.
  Provide the R code and the correlation values.
  (6 marks)

2. (b) Plot a scatter plot of `Sepal.Length` vs `Petal.Length` and include the correlation coefficient on the plot using the `text()` function in R.
   (4 marks)
3. (c) Write a brief interpretation of the correlation results from part (a).
   (2 marks)

END OF QUESTION PAPER

Key Points:

- Make sure students are able to demonstrate their understanding of R functions related to data manipulation, visualization, and statistical analysis.
- Each question tests a specific aspect of the syllabus, with a focus on practical R coding.
- The question paper covers dataframes, lists, matrices, arrays, reading files, extracting data, plots, regression, and correlation comprehensively.